

#	Risk(s) Identified	What might be affected and how	Existing controls & measures	Severity (a)	Likelihood (b)	Risk Rating (a x b)
1	Fingers trapped in gears or moving parts	User may get injured by moving mechanism	Enclose moving parts with protective casing; Limit motor torque/speed; Add emergency stop or power cutoff	4	3	12
2	Face tracking inaccuracies	System may lose track of the user or move incorrectly	Tune face detection parameters; Test in different lighting conditions; Implement smoothing or tracking persistence to reduce jitter	3	3	9
3	Voice command recognition error	Incorrect commands may be triggered	Use simple and distinct commands; Reduce background noise during operation; Provide fallback controls via the GUI	3	3	9
4	Time constraints	Project may not be fully completed or tested	Prioritise core functionality; Allocate time specifically for integration testing; Reduce non-essential features if necessary	3	3	9
5	Limited debugging time before demonstration	Unresolved bugs could appear during demo	Perform regular testing during development; Maintain modular code to simplify debugging; Prepare a simplified fallback demo if issues occur	3	3	9
6	Loose wiring or exposed connections	Risk of short circuit or minor electric shock	Insulate all connections; Secure wiring with cable management; Perform regular inspection before use	3	3	9
7	Arduino connection failure	System cannot communicate with motors or sensors	Test serial connection before operation; Implement error handling in MATLAB; Include a reconnect function in the GUI	4	2	8
8	Camera malfunction	Face tracking cannot operate correctly	Test camera initialisation at startup; Allow camera reset in software; Ensure drivers are installed and camera is securely connected	4	2	8
9	Motor failure	Tracker cannot physically move to follow the face	Test motors individually before integration; Ensure secure wiring and correct power supply; Include manual override or reset if motor stops responding	4	2	8
10	Poor lighting conditions	Face detection performance may degrade	Use consistent lighting during operation; Adjust camera exposure settings; Test system in expected environment	2	4	8
11	Multiple people in frame	System may track the wrong person	Configure system to prioritise largest or closest detected face; Restrict testing environment to a single user when possible	2	4	8
12	Noisy environment	Voice commands may not be recognised	Use quieter testing environment; Speak clearly and close to microphone; Provide alternative GUI controls if voice commands fail	2	4	8
13	Power supply misuse	Incorrect voltage may damage components or pose safety risk	Use regulated power supply; Label voltage requirements clearly; Include fuse or protection circuit	4	2	8
14	MATLAB GUI bugs	Buttons or controls may not perform intended actions	Test all GUI callbacks individually; Implement debugging messages; Test full workflow before demonstration	2	3	6
15	Serial communication delays	Slow response between MATLAB and Arduino	Limit frequency of serial messages; Buffer commands where possible; Optimise communication protocol	2	3	6
16	Different modules not integrating correctly	Face tracking, voice control, and GUI may conflict	Develop modules separately before integration; Integrate components step-by-step; Maintain clear interfaces between modules	2	3	6
17	Overheating components	Components may overheat causing burns or failure	Ensure adequate ventilation; Monitor temperature during operation; Avoid overloading circuits	3	2	6
18	Sharp edges on enclosure or components	Cuts or minor injuries during handling	Deburr edges after manufacturing; Use fillets in CAD design; Inspect before use	2	3	6
19	Unstable mounting or structure	Device may fall and cause injury or damage	Ensure stable base or mounting; Secure all components firmly; Test stability before operation	3	2	6